

THE GARDENING WRIST ORTHOTIC



DESIGN REQUIREMENTS

1. Performance
- Efficiency
2. Pain Reduction
3. Lower Risk of Injury
4. Cost-effective
5. Adaptive
6. Comfortability
7. Durability

COMPETITIVE ADVANTAGES

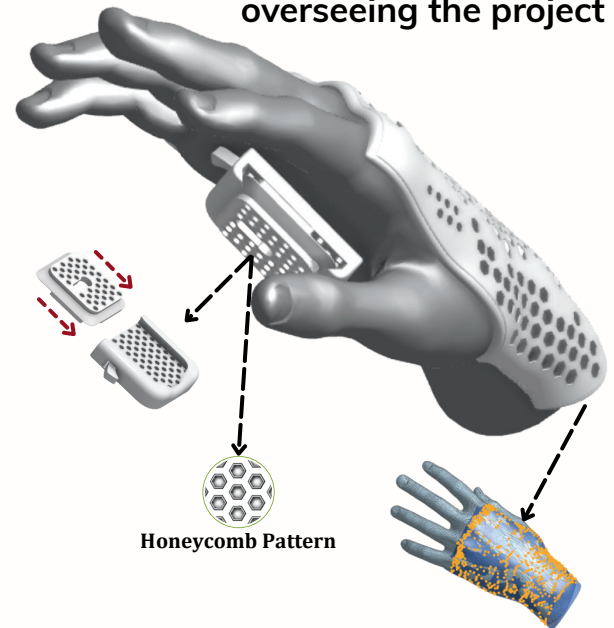
- Ultra-portable & Sturdy:
Weight 0.56 lbs (254 g), but
holds 25+ lbs (11.3+ kg)
- Highly cost-effective: Total
material cost of \$14.83
- Highly adaptable:
Compatible with most
common daily tools
- Personalized Fit: Custom-
molded Flexible brace using
your hand's mesh data
- Lightweight, Breathable
structure optimized for all-
day wear

Team N

Jonathan Yang
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Nick Farley
Dawson McMurry

Acknowledgements

Thank you Professor Jergens for
overseeing the project



VALUE PROPOSITION

- Equip time of less than
50 seconds
- Load bearing capacity
up to 10 lbs
- Prevents wrist over-
extension past 90
degrees
- Slot-clip mechanism
engages in 10-40
seconds
- Total weight under 1.5
lbs

Buckeye Engineering Design Showcase

DATE: April 28, 2026 **TIME:** 11AM - 3PM

LOCATION: Ohio Stadium & RPAC



THE OHIO STATE UNIVERSITY
COLLEGE OF ENGINEERING

Department of
ENGINEERING EDUCATION